One-day-ahead 1% VaR forecasting on QQQ daily log returns

Comparison of multiple models forecasting the left-tail conditional quantile of next-day log return

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Project Objective

Build and compare a reproducible VaR forecasting framework that combines:

- Rolling out-of-sample forecasts
- Multiple model classes (parametric, semi-parametric, nonparametric, direct quantile)
- Statistical backtesting
- Plot diagnostics

The real goal is not just in-sample fit. The goal is model comparison under realistic tail-risk forecasting conditions.

Data and sample

Dataset:

- QQQ daily adjusted close
- VIX close

Construction:

- Compute daily log returns from QQQ close orice: $r_t = \log\left(\frac{P_t}{P_{t-1}}\right)$
- VIX used as exogenous regressor for GARCHX
- Data pipeline starts at 2015-01-01

Current local files indicate:

- Full modeled sample runs roughly from 2015-01-02 to 2026-03-27
- Initial training fraction: 70%
- Out-of-sample period used in saved backtests: 848 trading days
- Pipeline forecasts start around 2022-11-01

Setup and notation

Let r_t denote the daily log return at time t. In several implementations, the code works with a scaled series $y_t = 100\, r_t$ for numerical stability when fitting volatility models.

For one-step-ahead Value-at-Risk at level α, the target object is the left-tail conditional quantile $\mathrm{VaR}_{t+1}(\alpha) = q_{t+1|t}(\alpha)$, so that ideally

$$\Pr(r_{t+1} < \mathrm{VaR}_{t+1}(\alpha) \mid \mathcal{F}_t) = \alpha.$$

Gaussian or Student-(t) tail

- If `dist="normal"`, then
$$q_\alpha = \Phi^{-1}(\alpha)$$
- If `dist="t"`, the code uses a Student-(t) quantile with estimated degrees of freedom (ν), then rescales it to unit variance by multiplying by
$$\sqrt{\frac{\nu-2}{\nu}}$$

When the model is conditionally location-scale, the generic structure is

$$r_{t+1} = \mu_{t+1} + \sigma_{t+1} z_{t+1},$$

which implies the final one-step forecast is

$$\mathrm{VaR}_{t+1}(\alpha) = \mu_{t+1} + \sigma_{t+1} q_\alpha,$$

where q_α is either a parametric quantile (Gaussian or Student-t) or an empirical residual quantile, μ is the mean comes from the fitted *arch_model(...).fit()* result, σ^2_{t+1} comes from the one-step-ahead conditional variance forecast through *arch_model.fit.forecast(horizon=1, reindex=False).variance*.

Model Suite

0. **Benchmark Model**
   - RiskMetrics with $\lambda = 0.94$ and initialized by variance estimate from previous 250 observations
1. **Original pipeline models**
   - GARCH(1,1) - Gaussian/Student-t
   - Nonparametric Quantile-GARCH SD - Gaussian/Student-t
   - Nonparametric-GARCH Bootstrap - Gaussian/Student-t
2. **Extension GARCH models**
   - GARCHX(VIX)
   - GJR-GARCH - Gaussian/Student-t - Raw/FHS

3. Direct Simulation Methods

- CAViaR-SAV
- Historical Simulation

* Financial returns usually have fatter tails than a Gaussian distribution. Gaussian model says extreme moves are very rare, but in real financial data, large losses and gains happen more often than Gaussian theory predicts. The Student-t distribution handles this better because it has heavier tails.

Model formulas / interpretation

RiskMetrics Model

$$r_t = \sigma_t W_t, W_t \sim N(0,1)$$
$$\sigma_t^2 = \lambda\sigma_{t-1}^2 + (1-\lambda)r_{t-1}^2$$
$$\mathrm{VaR}_{t+1}(\alpha) = \mu_{t+1} + \sigma_{t+1} q_\alpha,$$

GARCH(1,1)

Variance recursion:

$$rt = \mu + \varepsilon_t$$

$$\varepsilon_t = \sigma_t W_t, W_t \sim N(0,1)/Student - t$$
$$\sigma_t^2 = \omega + a\, r_{t-1}^2 + b\, \sigma_{t-1}^2,$$
$$\mathrm{VaR}_{t+1}(\alpha) = \mu_{t+1} + \sigma_{t+1} q_\alpha,$$

GARCHX(VIX)

GARCHX extends the standard GARCH volatility model by adding VIX as an exogenous market-risk variable in the conditional variance equation. The goal is to test whether implied market volatility helps explain the next-day tail risk of QQQ returns.

* R `garchx` package

$$\varepsilon_t = r_t - \mu$$

$$\varepsilon_t = \sigma_t W_t, W_t \sim N(0,1)$$

For one-step-ahead forecasting, the model uses the latest observed in-sample VIX value to form the next volatility forecast:

$$\sigma^2_{t+1|t} = \omega + \alpha\varepsilon_t^2 + \beta\sigma_t^2 + \delta x_t$$

where r_t is the QQQ daily log return, μ is the rolling sample mean, x_t is the standardized VIX regressor, σ_t^2 is the conditional variance.

The VaR forecast is then obtained by combining the rolling mean, the forecasted volatility, and a Gaussian tail quantile:

$$\widehat{\mathrm{VaR}}_{t+1}(\alpha) = \mu + \sigma_{t+1|t}\Phi^{-1}(\alpha)$$

GJR-GARCH

The GJR-GARCH(1,1,1) variance recursion is

$$\varepsilon_t = r_t - \mu$$
$$\sigma_t^2 = \omega + a\,\varepsilon_{t-1}^2 + \gamma\,\varepsilon_{t-1}^2 I(\varepsilon_{t-1} < 0) + b\,\sigma_{t-1}^2.$$

The indicator term makes negative shocks increase future volatility more than positive shocks of the same size when $\gamma > 0$.

The VaR mapping is the same as in GARCH:

$$\mathrm{VaR}_{t+1}(\alpha) = \mu_{t+1} + \sigma_{t+1} q_\alpha,$$

with either Gaussian or Student-t quantiles.

Why use it? Equity returns often display leverage/asymmetry: bad returns increase volatility more than good returns. GJR-GARCH is designed to capture that feature.

In the code, asymmetry is handled by arch_model(..., $p = 1$, $o = 1$, $q = 1$, ...).

Direct Quantile Forecasts

CAViaR-SAV

CAViaR models the quantile directly, instead of modeling volatility first.

The implementation here uses the **Symmetric Absolute Value (SAV)** form adopting the core idea from *CAViaR: Conditional Autoregressive Value at Risk by Regression Quantiles (Engle & Manganelli, 2002)*, moving attention from the full return distribution to the evolution of the tail quantile itself.

Symmetric Absolute Value (SAV) specification updates the quantile using its own lag and the magnitude of the lagged return.

In my implementation:

$$v_t = \beta_0 + \beta_1 v_{t-1} + \beta_2 |r_{t-1}|$$

and then maps it back to the left-tail return quantile by

$$q_t = -v_t.$$

So the one-step VaR forecast is simply the next autoregressive quantile level.

$$\widehat{\mathrm{VaR}}_{t+1}(\alpha) = q_{t+1}$$

The paper's SAV: the tail risk today depends on the previous tail level and on the size of the latest market move. Large absolute returns push the next quantile farther into the left tail.

Estimation criterion

The parameters are estimated by regression quantiles, using the standard quantile "check" loss:

$$L(\beta) = \sum_t \rho_\alpha(r_t - q_t(\beta))$$

where

$$\rho_\alpha(u) = u\,(\alpha - I(u < 0))$$

So CAViaR does not assume a Gaussian or Student-(t) conditional distribution. It estimates only the conditional quantile needed for VaR.

What the current simplifeid implementation does

In the code:

- returns are scaled by 100 before optimization
- the recursion is initialized from an empirical tail estimate
- only the **SAV** form is implemented
- optimization uses **Nelder–Mead only**
- only **4 starting values** are tried
- the iteration budget is light: `maxiter = 400`
- the final forecast is the next recursively computed quantile, then scaled back by dividing by 100

What aligns with the paper

- direct modeling of the conditional quantile rather than volatility-first modeling
- SAV structure driven by lagged quantile and lagged absolute return
- estimation by regression-quantile loss
- autoregressive tail-risk dynamics

What does not yet fully match the paper

- no **Dynamic Quantile (DQ) test** in the current project runner
- only the **SAV** variant is implemented; the paper also studies Adaptive, Asymmetric Slope, and Indirect GARCH CAViaR
- the optimizer is much weaker than the paper's search routine
- the paper uses a broader multi-start strategy and combines **Nelder–Mead with quasi-Newton refinement**
- initialization is simplified relative to the paper's empirical-quantile setup and broader estimation workflow.

Interpretation

In this project, the current CAViaR result is **exploratory**, because the backtest issue is now mainly **exception clustering and weak optimization**, not a broken recursion formula.

Historical Simulation

Rolling Historical Simulation is the simplest nonparametric benchmark. There is no explicit volatility model.

Given the most recent W returns, the forecast is just the rolling empirical quantile:

$$\mathrm{VaR}_{t+1}(\alpha) = \mathrm{Quantile}_{\alpha}(r_{t-W+1}, \dots, r_t).$$

In the code, the default window is

$$W = 250.$$

If fewer than 250 observations are available, the function uses the full available sample.

- **Strength**: model-free and easy to explain.
- **Weakness**: slow to adapt when volatility changes sharply.

It is still an important benchmark because many more complicated models do not always beat it.

Nonparametric Quantile-GARCH SD/Boostrap

Filtered Historical Simulation (FHS)

FHS fits a GARCH volatility model first, but instead of using a Gaussian or Student-(t) quantile at the end, it uses the **empirical quantile of standardized residuals**. So the forecast is

$$\widehat{\mathrm{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1}\hat{q}_{\alpha,\mathrm{emp}}$$

Lecture 18's **Nonparametric Quantile-GARCH SD**: forecast next-step volatility from GARCH, then combine it with the empirical residual tail. That is why FHS gives a more realistic left-tail fit than Gaussian GARCH in this project.

GARCH-FHS and GJR-FHS are semi-parametric models that use a parametric GARCH/GJR volatility fit and a nonparametric empirical residual quantile for VaR.

GARCH-FHS

Filtered Historical Simulation:

- fit GARCH
- compute standardized residuals
- take empirical alpha-quantile of residuals
- scale by $\hat{\sigma}_{t+1}$

GJR-FHS

Same empirical residual-quantile idea as FHS, but applied after fitting GJR-GARCH.

Residual Bootstrap GARCH

The bootstrap method is another semi-parametric VaR approach.

For the **one-step-ahead** VaR problem in this project, bootstrap and FHS are mechanically very close.

FHS computes

$$\widehat{\mathrm{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1}\hat{q}_{\alpha,\mathrm{emp}}$$

where $\hat{q}_{\alpha,\mathrm{emp}}$ is the empirical quantile of the standardized residuals.

Bootstrap instead resamples those same standardized residuals, forms

$$r^{*(i)}_{t+1} = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} z^{*(i)}_{t+1},$$

and then takes the empirical quantile across the simulated returns.

So for one-step forecasting, bootstrap is essentially just a simulation-based way to recover the same empirical residual tail used by FHS. That is why, in this project, bootstrap and FHS give very similar results.

GARCH-Bootstrap

One-step residual bootstrap:

- fit GARCH
- resample standardized residuals
- simulate one-step return
- take empirical alpha-quantile of simulated returns For one-step forecasting this is very close to FHS in practice.

Backtesting framework

At each out-of-sample date:

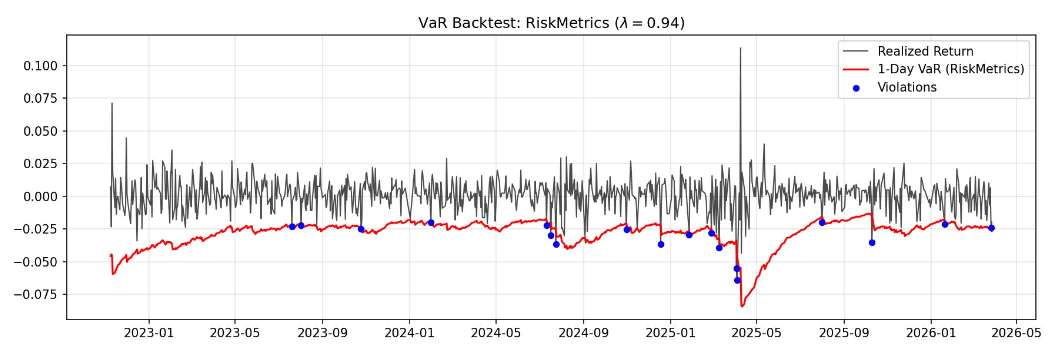
1. Fit model on data up to time t-1
2. Forecast one-step VaR for time t
3. Compare realized return r_t to forecast VaR_t
4. Record violation indicator:

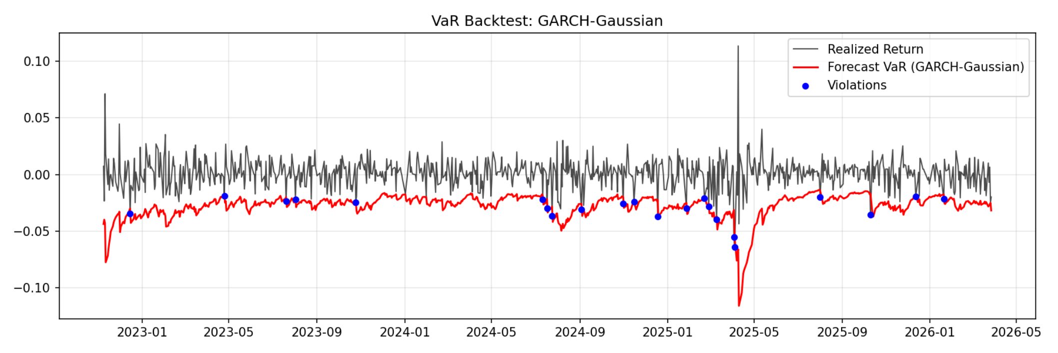
$$I_t = 1(r_t < VaR_t)$$

Backtests currently reported in the pipeline:

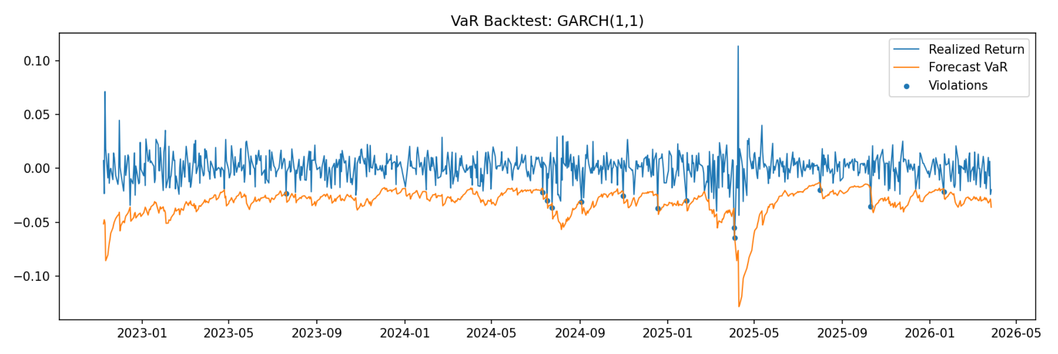
- Kupiec unconditional coverage test
- Christoffersen independence test
- Conditional coverage test

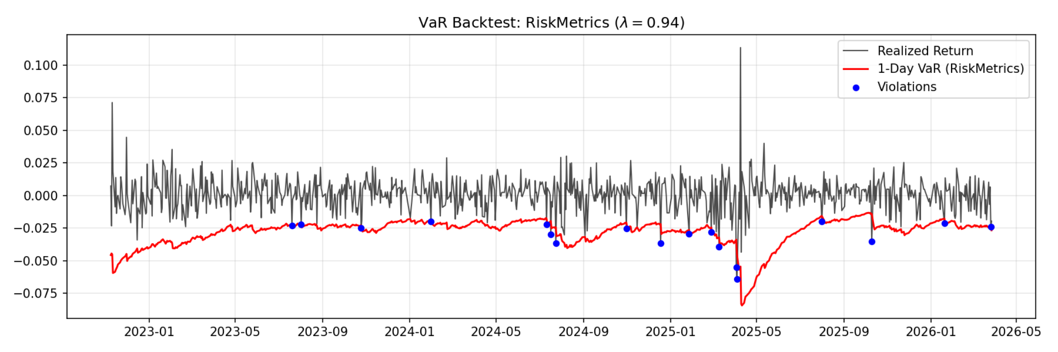
Plot Results

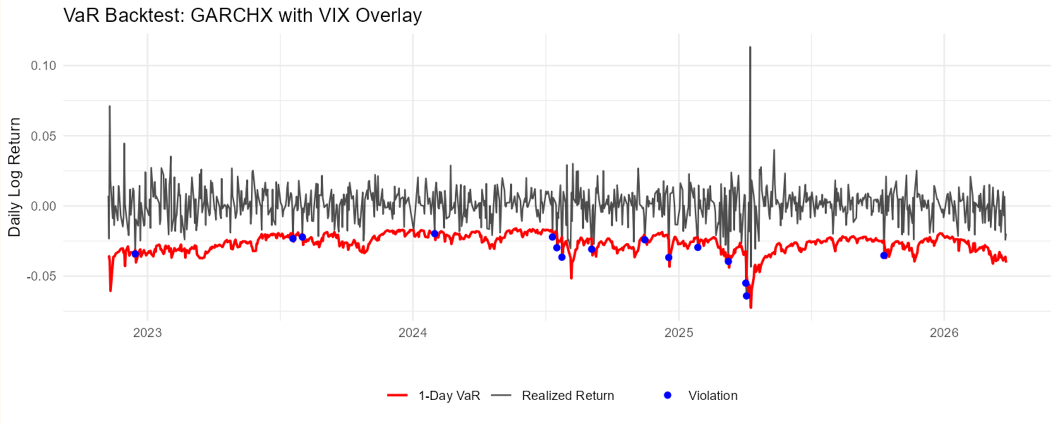



VaR Backtest: GARCH-Gaussian
Realized Return
Forecast VaR (GARCH-Gaussian)
Violations
0.10
0.05
0.00
−0.05
−0.10
2023-01
2023-05
2023-09
2024-01
2024-05
2024-09
2025-01
2025-05
2025-09
2026-01
2026-05

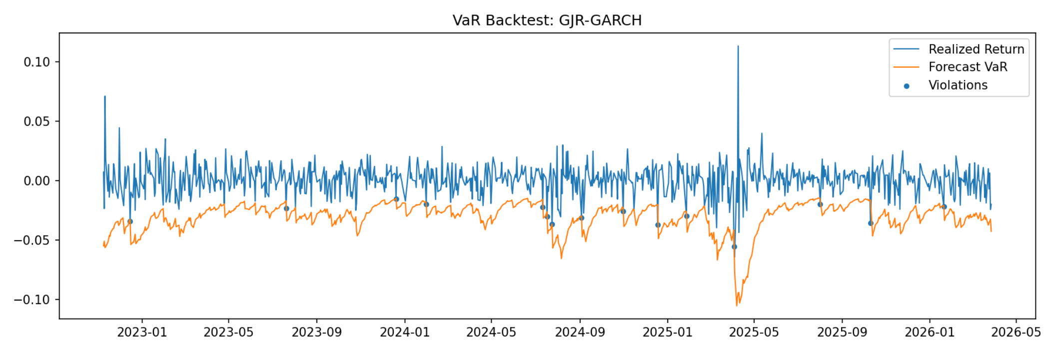
GARCH(1,1) - Student-t




VaR Backtest: RiskMetrics ($\lambda = 0.94$)
Realized Return
1-Day VaR (RiskMetrics)
Violations
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
2023-01
2023-05
2023-09
2024-01
2024-05
2024-09
2025-01
2025-05
2025-09
2026-01
2026-05

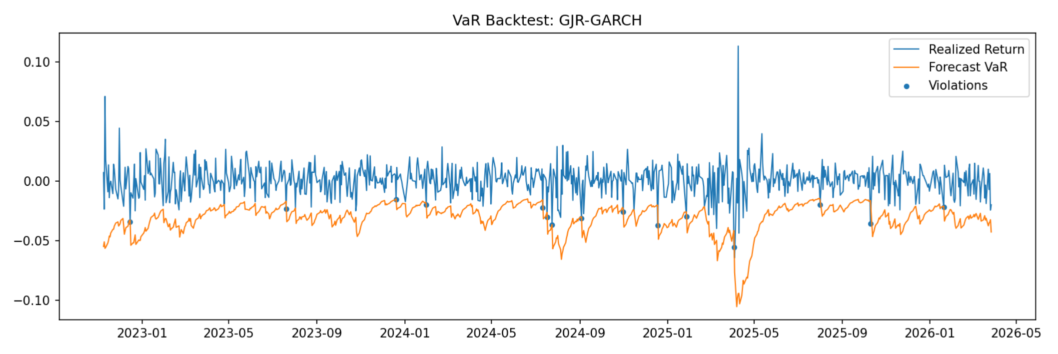


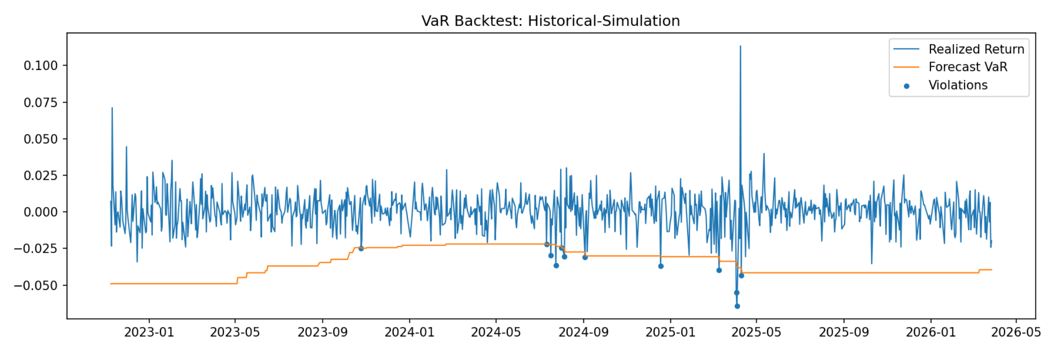
VaR Backtest: GARCHX with VIX Overlay
Daily Log Return
0.10
0.05
0.00
-0.05
2023
2024
2025
2026
1-Day VaR
Realized Return
Violation



VaR Backtest: GJR-GARCH
Realized Return
Forecast VaR
Violations
0.10
0.05
0.00
−0.05
−0.10
2023-01
2023-05
2023-09
2024-01
2024-05
2024-09
2025-01
2025-05
2025-09
2026-01
2026-05

GJR-GARCH - Student-t




VaR Backtest: Historical-Simulation
Realized Return
Forecast VaR
Violations
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
2023-01
2023-05
2023-09
2024-01
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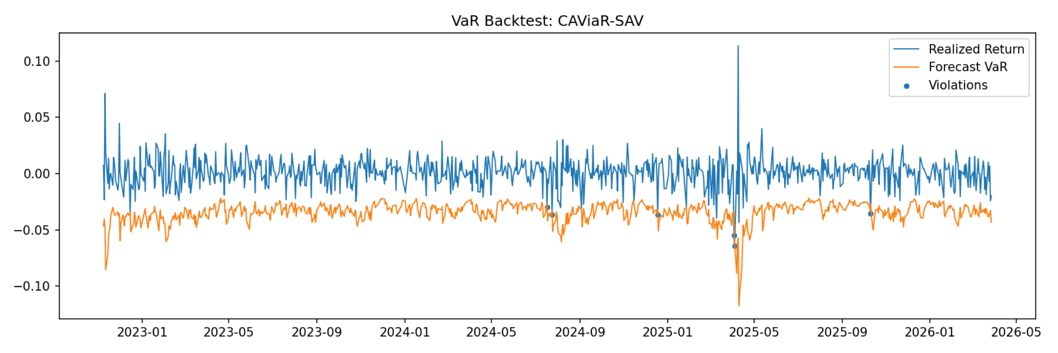
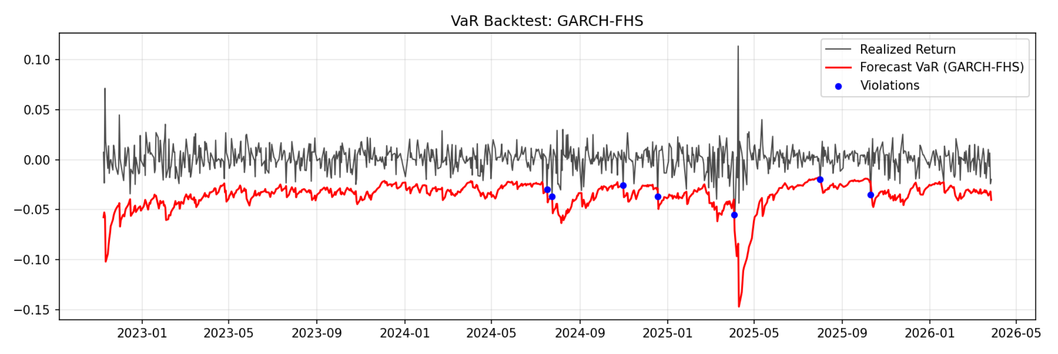
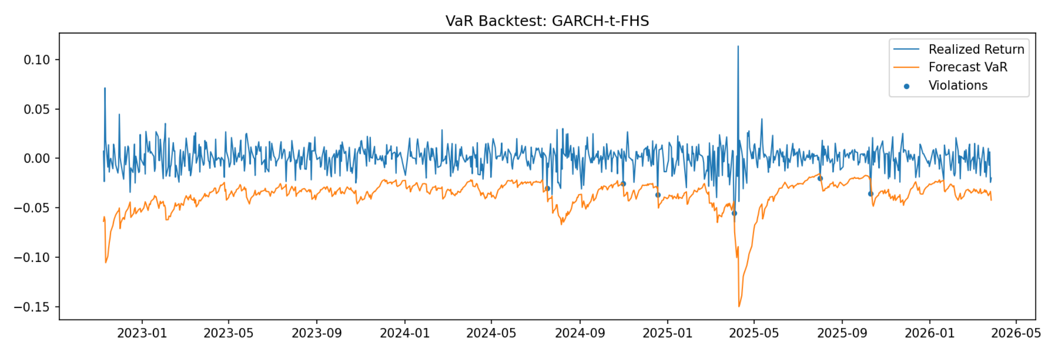
2025-01
2025-05
2025-09
2026-01
2026-05


VaR Backtest: CAViaR-SAV
Realized Return
Forecast VaR
Violations
0.10
0.05
0.00
−0.05
−0.10
2023-01
2023-05
2023-09
2024-01
2024-05
2024-09
2025-01
2025-05
2025-09
2026-01
2026-05

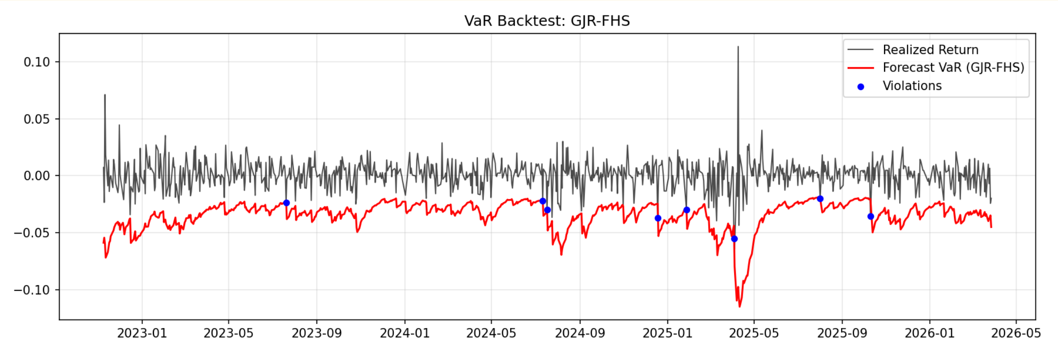
GARCH(1,1) - Gaussian - FHS



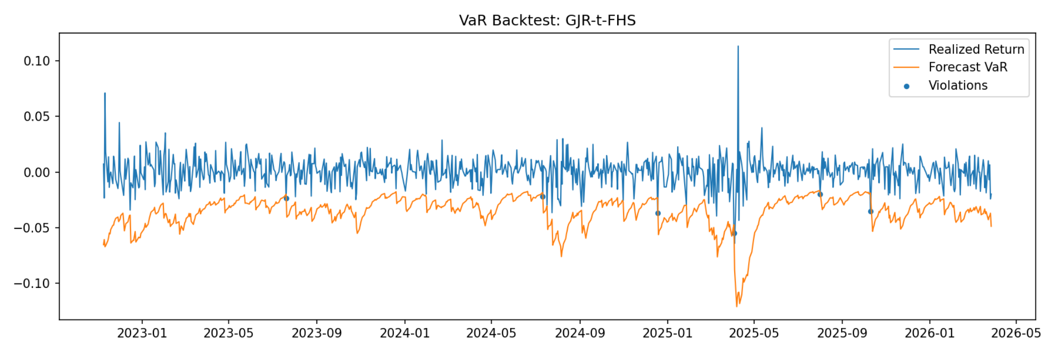
GARCH(1,1) - Student-t - FHS



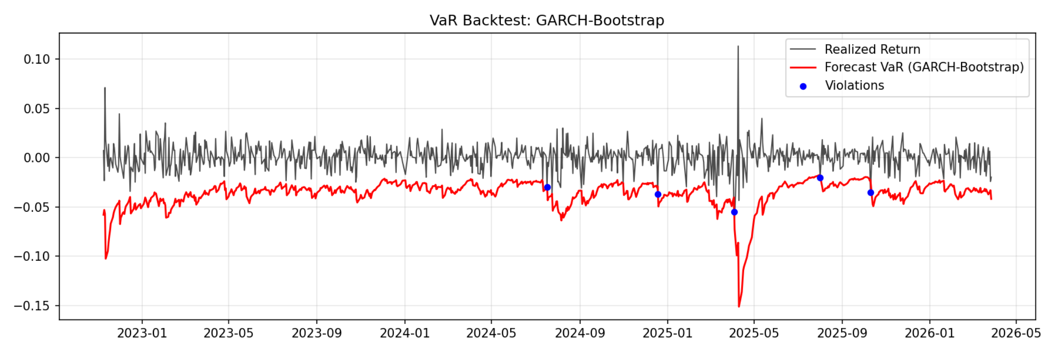
GJR-GARCH - Gaussian - FHS



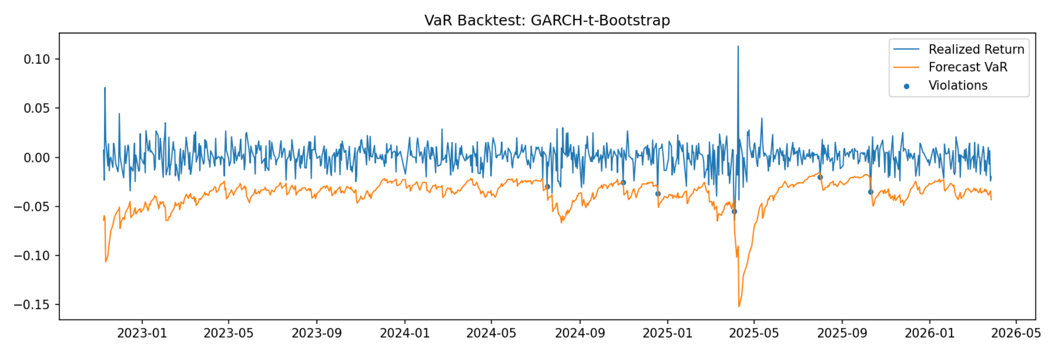
GJR-GARCH - Student-t - FHS



GARCH(1,1) - Gaussian - Bootstrap



GARCH(1,1) - Student-t - FHS



Plot-based interpretation

- Historical Simulation gives a stepwise, slow-moving VaR line but remains competitive
- GARCH Student-t reacts to stress periods reasonably well
- GJR-GARCH reacts to stress, but not as effectively as GJR-FHS
- GARCHX looks visually similar to plain GARCH but backtests worse
- GARCH-Gaussian sits too close to zero during stressed periods and gets breached too often
- GARCH-FHS and GJR-FHS become more negative during volatility spikes and produce much better calibration
- GARCH-Bootstrap is visually close to GARCH-FHS, which is expected for one-step forecasting
- CAViaR-SAV in the saved results still sits far too close to zero and gets breached constantly

Combined one-day backtest comparison

Rank	Model	Viol.	Rate	Kupiec p	Christoff. p	CC p	Interpretation
1	GJR-FHS	8	0.94%	0.867	0.696	0.914	Best overall calibration
2	GARCH-FHS	7	0.83%	0.598	0.733	0.821	Excellent, slightly conservative
3	GARCH-t-FHS	6	0.71%	0.366	0.770	0.637	Strong, conservative
4	GJR-t-FHS	6	0.71%	0.366	0.770	0.637	Strong, conservative
5	GARCH-t-Bootstrap	6	0.71%	0.366	0.770	0.637	Strong, deterministic bootstrap
6	Historical-Simulation	12	1.42%	0.253	0.157	0.191	Strong benchmark
7	GARCH-Student-t	13	1.53%	0.148	0.189	0.148	Acceptable baseline
8	CAViaR-SAV	6	0.71%	0.366	0.030	0.063	Coverage ok, clustering issue
9	GJR-GARCH	15	1.77%	0.042	0.462	0.097	Mixed, asymmetry not enough
10	GARCHX(VIX)	15	1.77%	0.042	0.262	0.068	Benchmark only
11	RiskMetrics (0.94)	18	2.12%	0.004	0.392	0.012	Underestimates risk
12	GARCH-Gaussian	22	2.59%	0.0001	0.597	0.0005	Too thin-tailed

Interpretation:

- Too many violations -> model underestimates risk
- Too few violations -> model is conservative / overestimates risk
- Failure of independence -> violations cluster, model misses dynamic risk structure

Overall performance evaluation

Outperformers

- **GJR-FHS** is the best final model: hit rate closest to target and strongest joint backtesting results
- **GARCH-FHS** is a very close second and the cleanest simple recommendation
- **Student-t FHS / bootstrap variants** also perform very well, but are slightly more conservative
- **Historical Simulation** remains a strong benchmark despite its simplicity

Models that do not work as well

- **GARCH-Gaussian** fails clearly: too many violations, so the Gaussian tail is too thin for QQQ downside risk
- **RiskMetrics** also underestimates tail risk and is dominated by the FHS models
- **GARCHX(VIX)** is useful as an exogenous-volatility benchmark, but it does not improve forecasting accuracy in the saved outputs

Conclusion

- The biggest improvement comes from **empirical-tail calibration**, not just from changing the volatility recursion
- Asymmetry helps, but **good tail modeling helps more**
- For QQQ 1% VaR, volatility models with **empirical residual tails** outperform purely parametric Gaussian approaches

Project limitations

- **CAViaR is only a partial replication:** It uses simplified optimization and omits the paper's full Dynamic Quantile adequacy testing, so its results are not fully validated.
- **GARCHX is not fully justified yet:** VIX is included as an exogenous regressor, but the project does not formally verify that it adds forecasting value beyond plain GARCH.
- **Some implementation choices may bias interpretation:** Deterministic bootstrap seeding and split model runners can make comparisons look cleaner or more stable than they truly are.

References

Engle, R. F., & Manganelli, S. (2002, July). CAViaR: Conditional autoregressive value at risk by regression quantiles [Unpublished manuscript].

Hansen, P. R., & Lunde, A. (2001, March 8). A comparison of volatility models: Does anything beat a GARCH(1,1)? (Working Paper Series No. 84).

Ratih, I. D., Ulama, B. S. S., & Prastuti, M. (2018). Value-at-risk analysis using ARMAX GARCHX approach for estimating risk of banking subsector stock return's. IOP Conference Series: Journal of Physics: Conference Series, 974, 012029. doi:10.1088/1742-6596/974/1/012029